

Heatwave report card — May 2024 heatwave

REPLAY

SYNTHETIC REPORTS

SIMULATED DELIVERY

Ahmedabad, 48 wards · event 16 May – 30 May 2024 (15 days, peak 26 May) · model estimates, not clinical predictions · event 1 of 2

48 / 48

wards reached Red (mortality risk)

344

Red ward-days · 680 Orange-or-worse

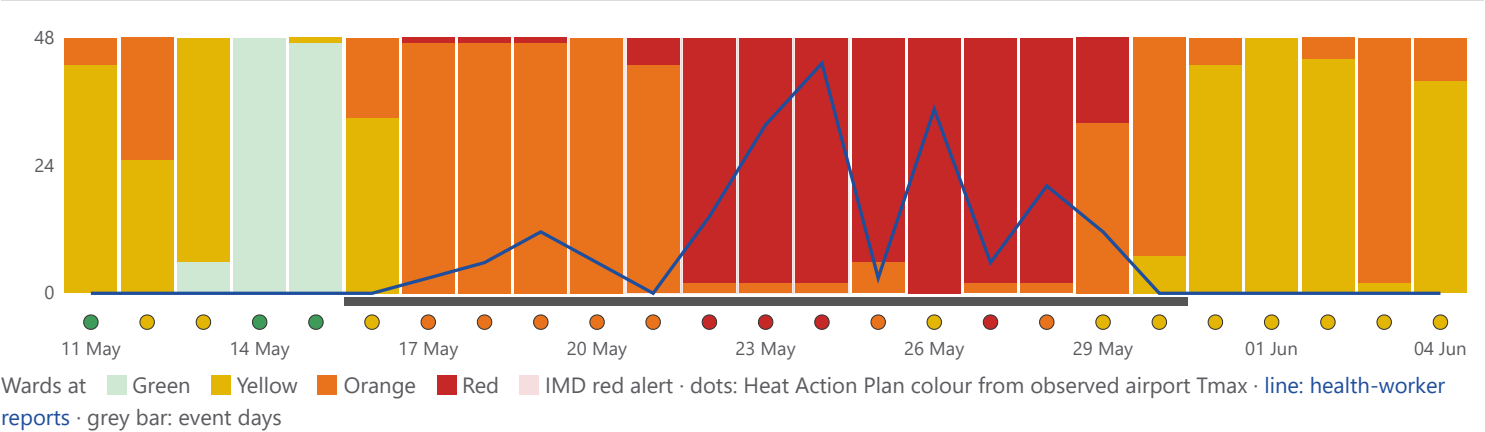
238 of 344

Red ward-days with P(Red) ≥ 40% in forecasts 3–5 days ahead

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Red ward-days covered by an approved Red alert

What happened, day by day



Checks against observations

IMD red-alert days (20 May–24 May) with a ward at Red	4 of 5
Days of Orange-or-worse warning before IMD's red alert	4
Plan Orange/Red days (observed Tmax) the model also rated Orange+ (median ward)	11 of 11
Missed: plan Orange/Red, model below	0
Over-warning: model Orange+, plan Yellow or none	4 of 14
Same level as the plan	18 of 25 days

References: IMD red alert for Ahmedabad 20–24 May 2024; Heat Action Plan colour from observed Tmax at the airport (IMD 42647). Health outcomes are city-level only (69 heatstroke cases in May 2024).

Health-worker reports

66 reports in the window (synthetic demo data). 3 ward-days flagged for more cases than the model's share.

Ward	Day	Reports	Expected	MRI
Vatva	23 May	6	0.2	90.7
Maktampura	24 May	3	0.25	86.3
Vatva	24 May	7	0.27	91.8

Alerts and actions

32 alert drafts: 25 dispatched, 7 draft.				
Messages: whatsapp 75 simulated.				
Level	Action	Alerts	Days	Officer
Red	approved	11	18 May–29 May	Demo officer (simulated)
Red	dispatched	11	18 May–29 May	Demo officer (simulated)
Orange	approved	14	16 May–30 May	Demo officer (simulated)
Orange	dispatched	14	16 May–30 May	Demo officer (simulated)

Every decision is in the audit log with officer name and time.

Most affected wards

Ward	Red days	Peak MRI	Reports / exp.
Baherampura	12	100	0 / 2.0
Indrapuri	9	99	0 / 1.3
Shahpur	9	98	2 / 1.2
Asarwa	9	98	0 / 0.9
Danilimda	9	98	2 / 1.1
Saraspur-Rakhiyal	8	96	0 / 1.2
Lambha	8	96	2 / 2.9

Ranip	8	96	0 / 1.2
Odhav	8	96	3 / 1.9
Bhaipura Hatkeshwar	8	95	0 / 0.8

Limitations

- Model estimates, not clinical predictions. Health outcomes for this event are known only city-wide.
- Early warning in a replay uses archived deterministic forecasts from three models (9 members), not the live 122-member ensemble.
- Sheet-roof shares are census-based estimates; children under 5 are census 2011 constituency averages; elderly and outdoor-worker indicators are held at the city midpoint.
- Report flags are prompts to verify, not confirmed clusters: at $p < 0.01$ over hundreds of ward-days, about one flag can occur by chance.
- Health-worker reports in this card are SYNTHETIC demo data.
- Alert deliveries were SIMULATED: nothing was sent to real phones.

Appendix — all wards

Ward	Red days	Orange+ days	First Red	Red days alerted	Peak MRI	PVI	Reports	Expected
Baherampura	12	15	17 May	11	100	66	0	2.0
Indrapuri	9	15	21 May	9	99	57	0	1.3
Shahpur	9	15	21 May	9	98	56	2	1.2
Asarwa	9	15	21 May	9	98	56	0	0.9
Danilimda	9	15	21 May	9	98	56	2	1.1
Saraspur-Rakhiyal	8	15	22 May	8	96	56	0	1.2
Lambha	8	15	22 May	8	96	59	2	2.9
Ranip	8	15	22 May	8	96	55	0	1.2
Odhav	8	15	22 May	8	96	54	3	1.9
Bhaipura Hatkeshwar	8	15	22 May	8	95	53	0	0.8
Amraiwadi	8	15	22 May	8	95	54	2	1.3
Sardarnagar	8	15	22 May	8	94	55	2	1.4
Ghatlodia	8	15	22 May	8	94	53	0	1.0
Nikol	8	14	22 May	8	94	54	0	2.3
India Colony	8	15	22 May	8	94	53	0	1.6
Thakkarbapanagar	8	15	22 May	8	93	51	1	1.0
Vatva	7	14	22 May	7	93	53	13	1.2
Chandlodia	7	14	22 May	7	93	53	1	0.7
Sabarmati	7	14	22 May	7	93	51	1	1.4
Vejalpur	7	14	22 May	7	93	51	2	1.0
Vasna	7	14	22 May	7	92	50	0	1.2
Jamalpur	7	14	22 May	7	92	48	0	0.9
Gota	7	14	22 May	7	91	52	3	1.1
Sarkhej	7	14	22 May	7	91	53	2	2.4
New Wadaj	7	14	22 May	7	91	45	0	0.9
Ramol Hathijan	7	14	22 May	7	91	52	2	2.3
Bodakdev	7	14	22 May	7	91	50	2	1.1
Vastral	7	14	22 May	7	91	51	2	0.9
Saijpur Bogha	7	14	22 May	7	91	49	2	1.1
Gomtipur	7	14	22 May	7	91	49	2	1.2
Naroda	7	14	22 May	7	91	50	2	2.5
Thaltej	7	14	22 May	7	91	51	1	1.1
Khadia	7	14	22 May	7	90	48	3	1.2
Shahibag	7	14	22 May	7	90	47	2	1.3
Bapunagar	7	14	22 May	7	90	47	1	1.0
S.P.Stadium	7	14	22 May	7	89	46	0	1.4
Maninagar	7	14	22 May	7	89	46	0	1.7
Viratnagar	7	14	22 May	7	89	47	1	1.4

Isanpur	7	14	22 May	7	89	46	0	1.4
Kubernagr	7	14	22 May	7	89	46	0	0.5
Maktampura	7	13	22 May	7	88	49	3	1.1
Khokhra	7	14	22 May	7	87	44	0	1.3
Dariyapur	6	13	22 May	6	86	42	1	0.6
Chandkheda	6	13	22 May	6	86	44	0	0.4
Jodhpur	6	13	22 May	6	84	42	1	2.1
Navrangpura	6	13	22 May	6	84	40	2	3.5
Paldi	1	13	26 May	1	81	35	0	1.6
Naranpura	1	13	26 May	1	81	35	3	1.5

Early-warning source: Weather: ERA5 reanalysis. Probabilities: forecasts issued 3-5 days ahead by ECMWF, GFS and ICON (9 members), i.e. what the system would have shown 3 days before each day.